

# PromptShala

Practical AI Literacy Studio for Educators

Version 1.2 · Revised per client feedback (21 & 28 Aug 2026)

|                 |                                                                       |
|-----------------|-----------------------------------------------------------------------|
| Course          | Educational Application Development — ET 617                          |
| Client          | Centre for Educational Technology (CET), IIT Bombay                   |
| Project Leads   | Prof. Ramkumar Rajendran & Aditya                                     |
| Student Team    | Darshan Sonawane · Vishal Patel · Ashok Kumar · Raghuram Gundi        |
| Version / Date  | Version 1.2   2 September 2026                                        |
| Document Status | Draft (Revision 2) — incorporates client review and recovery schedule |

**What changed in v1.2 (client feedback, 21 & 28 Aug 2026):**

- Product name finalized as **PromptShala** — Practical AI Literacy Studio for Educators.
- Primary teacher persona expanded with classroom jobs, constraints, risks, behaviour and feature rationale.
- Official source repository recorded as [github.com/EAD-Labs/Group-6\\_2026](https://github.com/EAD-Labs/Group-6_2026); GitHub and Jira working rules added.
- Milestones replaced with a **week-wise recovery and delivery schedule** following the 28 Aug review.
- Each delivery module retains separate staging, testing, evidence and client review.
- The complete project presentation remains fixed for **15 October 2026**.
- The signed Version 1.1 page is preserved unchanged as the final approval-record page.

Prepared as the agreed design basis for a teacher-centred, safe and scalable AI-literacy experience.  
Reference: [ams.iitb.ac.in/d/276931](https://ams.iitb.ac.in/d/276931) · [ams.iitb.ac.in/d/276937](https://ams.iitb.ac.in/d/276937)

## 1. Project Information

|                        |                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title          | PromptShala — Practical AI Literacy Studio for Educators                                                                                                  |
| Client                 | Centre for Educational Technology (CET), IIT Bombay                                                                                                       |
| Client Representatives | Prof. Ramkumar Rajendran and Aditya                                                                                                                       |
| Client Contact         | ramkumar.rajendran@iitb.ac.in                                                                                                                             |
| Student Team           | Darshan Sonawane (23b1541); Vishal Patel (23b0745); Ashok Kumar (23b0628); Raghuram Gundi (23b1534)                                                       |
| Primary Users          | School and higher-education teachers, teacher educators, and professional-development participants                                                        |
| Application Type       | Responsive web application; progressive-web-app compatible                                                                                                |
| Target Platform        | Modern desktop, laptop, tablet and mobile web browsers                                                                                                    |
| Official Repository    | <a href="https://github.com/EAD-Labs/Group-6_2026">https://github.com/EAD-Labs/Group-6_2026</a> (created; initial project setup pending as of 2 Sep 2026) |
| HLD Version            | 1.2 (Revision 2)                                                                                                                                          |
| Date / Status          | 2 September 2026   Draft for consolidated client review                                                                                                   |

### 1.1 Document Purpose

This HLD converts the client project brief into a development-ready design baseline. It defines users, workflows, architecture, data handling, scope, measurable acceptance criteria, responsibilities and the milestone agreement for the ET 617 student project.

**Revision rule:** Version 1.2 incorporates the client directions recorded on 21 and 28 Aug. Any new scope or timeline change must be entered in the decision log and reflected in Jira. The original signed Version 1.1 page is retained unchanged as an approval-history record.

## 2. Project Overview

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### 2.1 Problem Statement

Teacher professional-development programmes often introduce Generative AI through isolated workshops or demonstrations. Teachers may understand that AI tools exist but still lack sustained practice in structuring prompts, checking AI-generated output, creating reusable workflows, and applying AI responsibly to authentic classroom tasks. Without guided practice and formative feedback, adoption remains inconsistent and dependent on individual experimentation.

### 2.2 Proposed Solution

PromptShala is a self-paced, interactive professional-development platform that combines concise concept learning with guided experimentation. Participants progress through **four sequenced learning modules** covering Generative-AI foundations, prompt engineering, reusable AI assistants, and AI-assisted transformation of teacher-owned learning materials. The platform provides quizzes, an AI-supported prompt sandbox, structured feedback, progress tracking, controlled module unlocking and a completion certificate.

**Design principle:** Learning is demonstrated through authentic teaching work — prompts, assistants and transformed resources — rather than passive content completion alone.

**Note on terminology:** the *learning modules* above are course content for participants. They are distinct from the *three delivery modules* used to build and test the software (see Sections 13 and 17.4).

### 2.3 Project Objectives

1. Build practical teacher competency in Generative-AI concepts, limitations, responsible use, APIs and prompting.
2. Develop participants' ability to create clear, contextual and testable prompts through repeated practice and formative feedback.
3. Enable teachers to design reusable AI assistants and repeatable workflows for lesson planning, assessment and resource creation.
4. Help teachers transform their own instructional materials into summaries, quizzes, study guides and related classroom resources while applying privacy and human-review safeguards.
5. Provide transparent progress, module-gating and completion evidence that supports individual learning and programme monitoring.

### 2.4 Deployment Context

**Intended context (revised):** Professional development for **school teachers and other educators**. PromptShala is delivered as self-paced online learning together with facilitated teacher-development workshops, and is intended for educators across schools and institutions — not for a single campus. CET / IIT Bombay is the commissioning client and pilot coordinator; the learning experience itself is aimed at school teachers and educators generally.

**Expected users (revised):** Proposed **initial pilot of 10 participants** (school teachers and educators). The architecture is designed for up to **25 concurrent authenticated users** to provide comfortable headroom during facilitated sessions and future cohorts.

**Technology context:** Participants may use personal or institutional laptops, tablets or phones over broadband or 4G. Core reading can tolerate intermittent connectivity; AI practice and authenticated progress updates require internet access.

### 3. User Personas

Persona note: The personas are fictional composites grounded in the client discussions of 7, 14, 21 and 28 Aug 2026. They describe design requirements, not research findings.

#### 3.1 Primary Persona — Meera, Beginner School Teacher

|                       |                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teaching context      | Teacher of Classes 5–10, commonly Mathematics or Science; 5–15 years of experience; 35–45 learners per class; limited preparation time.                     |
| Current AI experience | Beginner. May have tried ChatGPT or Gemini for a question or worksheet, but does not consistently define role, learner, context, outcome or output format.  |
| Digital context       | Comfortable with WhatsApp, documents, presentations and web forms; uses a personal Android phone and sometimes a laptop; connectivity may vary.             |
| Motivation            | Save preparation time and create age-appropriate lessons, explanations, assessments, slides, parent communication and differentiated resources.             |
| Main anxiety          | Giving wrong information to learners, exposing student information, copying copyrighted content, or appearing unskilled when the tool produces poor output. |
| Time available        | Short 20–30 minute learning sessions; needs immediate practice and visible improvement rather than long theory-first instruction.                           |

| Teacher job               | Current difficulty                                                                   | PromptShala support                                                                                         |
|---------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Explain a concept         | Generic AI explanations ignore grade, prior knowledge and misconceptions.            | Scenario-based prompt practice requires learner level, teaching role, context and expected format.          |
| Create assessments        | Questions may be poorly levelled, repetitive or disconnected from learning outcomes. | Guided prompts cover outcome, Bloom level, question type, answer key and review checklist.                  |
| Prepare lessons/resources | Repeatedly rewrites similar instructions for worksheets, slides and summaries.       | Gem-style assistant builder turns a validated prompt into a reusable teaching workflow.                     |
| Use own materials         | Uncertain how to safely convert notes or a syllabus into useful resources.           | NotebookLM walkthrough and content-transformation safeguards emphasise source grounding and no student PII. |

#### Behaviour, Barriers and Design Response

| Observed / Expected Behaviour                                      | Design Requirement                                                                     | Feature Evidence                                                                |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Starts with a vague one-line request and accepts the first output. | Show what is missing without technical jargon and invite revision.                     | Visible rubric, dimension feedback and side-by-side attempt comparison.         |
| Prefers doing a classroom task to reading AI theory.               | Introduce AI at a high level through questions, examples and choices.                  | Interactive onboarding and authentic scenarios before optional readings/videos. |
| May be uncomfortable entering an API key.                          | Delay API setup until needed; explain purpose, local handling and non-storage clearly. | Guided API-key setup, consent message and provider-independent fallback.        |
| Uses a phone and may lose connectivity.                            | Short screens, saved drafts, clear resume state and no horizontal scrolling.           | Responsive UI, autosave, progress persistence and retry-safe error states.      |

#### Why This Persona Is Different from a Generic Professional User

| Dimension         | School Teacher                                                                                                 | Banker / Architect / Generic Professional                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Primary recipient | Children with varied age, prior knowledge and learning needs.                                                  | Adult clients, colleagues, regulators or technical stakeholders.                           |
| Typical output    | Lesson plan, explanation, quiz, rubric, worksheet, parent message, classroom slides.                           | Financial analysis, compliance note, design specification, proposal or operational report. |
| Quality test      | Age appropriateness, pedagogy, curriculum fit, accessibility, misconception handling and teacher verification. | Commercial, regulatory, technical or client-specific correctness.                          |
| Safety boundary   | No identifiable student data; teacher remains responsible for every classroom use.                             | Domain confidentiality and professional controls, but not child-centred pedagogy.          |

**Primary success criterion (20–30 minute experience):** Meera can write and improve one classroom prompt using the agreed LEARN-LM / CRAFT-style dimensions, explain what was missing, save a reusable Gem-style specification, identify when NotebookLM is useful, and state the privacy and human-review safeguards before using the output with learners.

### 3.2 Secondary Persona — Anil, Teacher Educator / Facilitator

Anil facilitates teacher-development programmes. He needs a consistent sequence, aggregate cohort progress, evidence of practice, common-misconception summaries and clear intervention points without routine access to private prompts or uploaded materials.

### 3.3 Supporting Roles

| Persona                 | Primary Goal                                           | Main Need                                                  |
|-------------------------|--------------------------------------------------------|------------------------------------------------------------|
| CET Content Curator     | Maintain accurate, sequenced teacher-learning content. | Draft, preview, publish and version controls.              |
| Programme Administrator | Operate cohorts and protect access.                    | User, role, certificate, audit and configuration controls. |
| Institutional Sponsor   | Understand adoption and completion.                    | Aggregated, privacy-preserving programme reports.          |

## 4. User Roles & Permissions

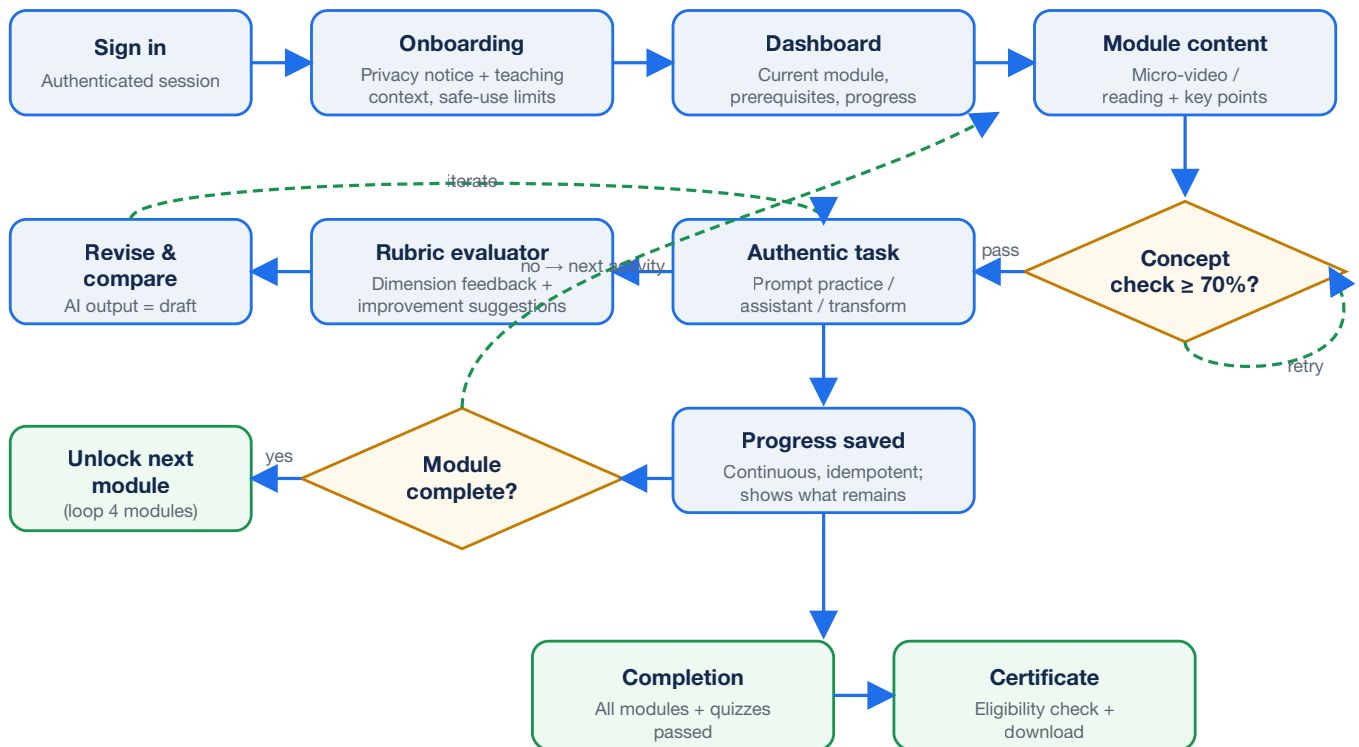
The product uses role-based access control. 'Participant' refers to the teacher or teacher educator completing the course; the Student Team is a development team and is not an application user role.

**Permission safeguard:** Private prompts and uploaded materials are not exposed to facilitators or administrators for routine analytics. Any exceptional support access must be explicit, time-limited and audited.

| Capability                          | Participant            | Facilitator / Content Manager         | Administrator                     |
|-------------------------------------|------------------------|---------------------------------------|-----------------------------------|
| Authenticate and manage own profile | Own account            | Own account                           | All accounts                      |
| View published learning content     | Allowed                | Allowed + preview drafts              | Allowed + preview drafts          |
| Complete quizzes and practice       | Own attempts           | Optional test mode                    | Optional test mode                |
| Use prompt sandbox / AI tools       | Own workspace          | Own workspace + examples              | Own workspace + policy controls   |
| View progress                       | Own progress           | Assigned cohort; aggregate by default | All cohorts; aggregate by default |
| Create or edit learning content     | Not allowed            | Draft and edit                        | Draft, edit and publish           |
| Manage module prerequisites         | Not allowed            | Recommend                             | Configure                         |
| Manage users and roles              | Not allowed            | Not allowed                           | Allowed                           |
| Issue / revoke certificates         | Download own           | Recommend                             | Allowed with audit record         |
| Export reports                      | Own completion summary | Assigned cohort                       | Programme-level                   |
| View raw private uploads            | Own files only         | Not allowed by default                | Break-glass support only; audited |

## 5. User Journeys & Core Workflows

### 5.1 Primary User Journey



**Figure 1.** Primary participant journey — corrected to show the prompt-practice feedback loop, the quiz-gated module unlock ( $\geq 70\%$ ), and completion  $\rightarrow$  certificate.

A participant signs in, completes a short onboarding and sees a dashboard showing the current module, prerequisites and progress. Each module combines a micro-video or reading with a concept check and an authentic task. The system saves progress continuously and explains what remains before the next module unlocks.

During prompt practice, the participant submits a prompt to a rubric-driven evaluator. The platform returns dimension-level feedback, an explanation and improvement suggestions; the participant can revise and compare attempts. AI output is presented as a draft requiring teacher review, not as an authoritative classroom answer.

### 5.2 Secondary User Journey

1. Facilitator or content manager creates or revises a learning item in Draft status.
2. The item is previewed across desktop and mobile layouts and checked for accessibility.
3. An administrator approves publication and assigns the item to the appropriate module / cohort.
4. The facilitator monitors aggregated completion, quiz patterns and intervention flags.
5. The team revises content based on evidence while preserving historical attempt records.
6. At programme completion, authorized staff export a cohort summary and certificate register.

### 5.3 Key Use Cases

| ID    | User        | Description                                         | Expected Outcome                                                                   |
|-------|-------------|-----------------------------------------------------|------------------------------------------------------------------------------------|
| UC-01 | Participant | Sign in and resume from the last completed activity | Authenticated session and accurate progress state                                  |
| UC-02 | Participant | Complete a lesson and auto-graded quiz              | Immediate explanation, retry option and prerequisite update                        |
| UC-03 | Participant | Submit and revise a prompt in the sandbox           | Dimension scores, feedback and attempt comparison                                  |
| UC-04 | Participant | Configure a reusable teaching-assistant template    | Saved assistant specification with purpose, context, constraints and output format |

|       |               |                                               |                                                                                    |
|-------|---------------|-----------------------------------------------|------------------------------------------------------------------------------------|
| UC-05 | Participant   | Transform teacher-owned material              | Editable draft summary, quiz or study guide with review checklist                  |
| UC-06 | Facilitator   | Review assigned-cohort progress               | Aggregate completion and misconception indicators without exposing private content |
| UC-07 | Administrator | Publish content and issue completion evidence | Versioned content and auditable certificate record                                 |



## 6. Functional Architecture

### 6.1 Major System Modules

| ID | Module                    | Purpose and Main Functionality                                                               |
|----|---------------------------|----------------------------------------------------------------------------------------------|
| M1 | Identity & Access         | Authentication, participant profile, roles, consent and secure session management            |
| M2 | Learning Experience       | Four-module pathway, micro-videos, readings, examples and activity navigation                |
| M3 | Assessment & Progress     | Quizzes, attempts, explanations, prerequisite rules, progress and completion state           |
| M4 | Prompt Practice           | Guided sandbox, rubric evaluation, improvement suggestions and attempt comparison            |
| M5 | AI Workflow Tools         | Reusable assistant specification, content transformation and NotebookLM workflow guide       |
| M6 | Certification             | Eligibility checks, certificate generation, download and certificate register                |
| M7 | Administration & Insights | Content lifecycle, users / roles, cohorts, aggregate reports, configuration and audit review |

### 6.2 Module Dependency

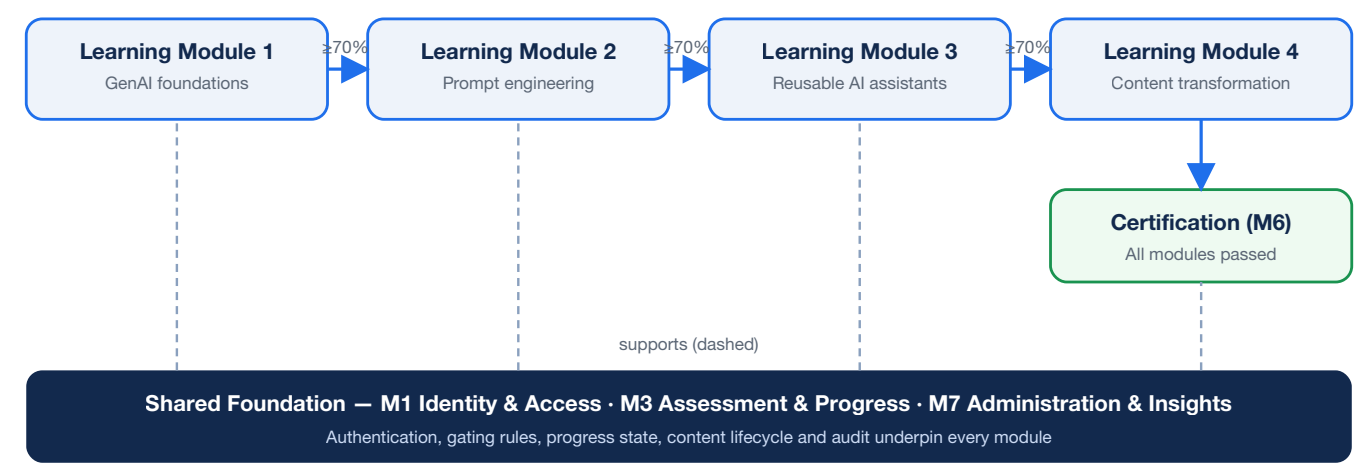
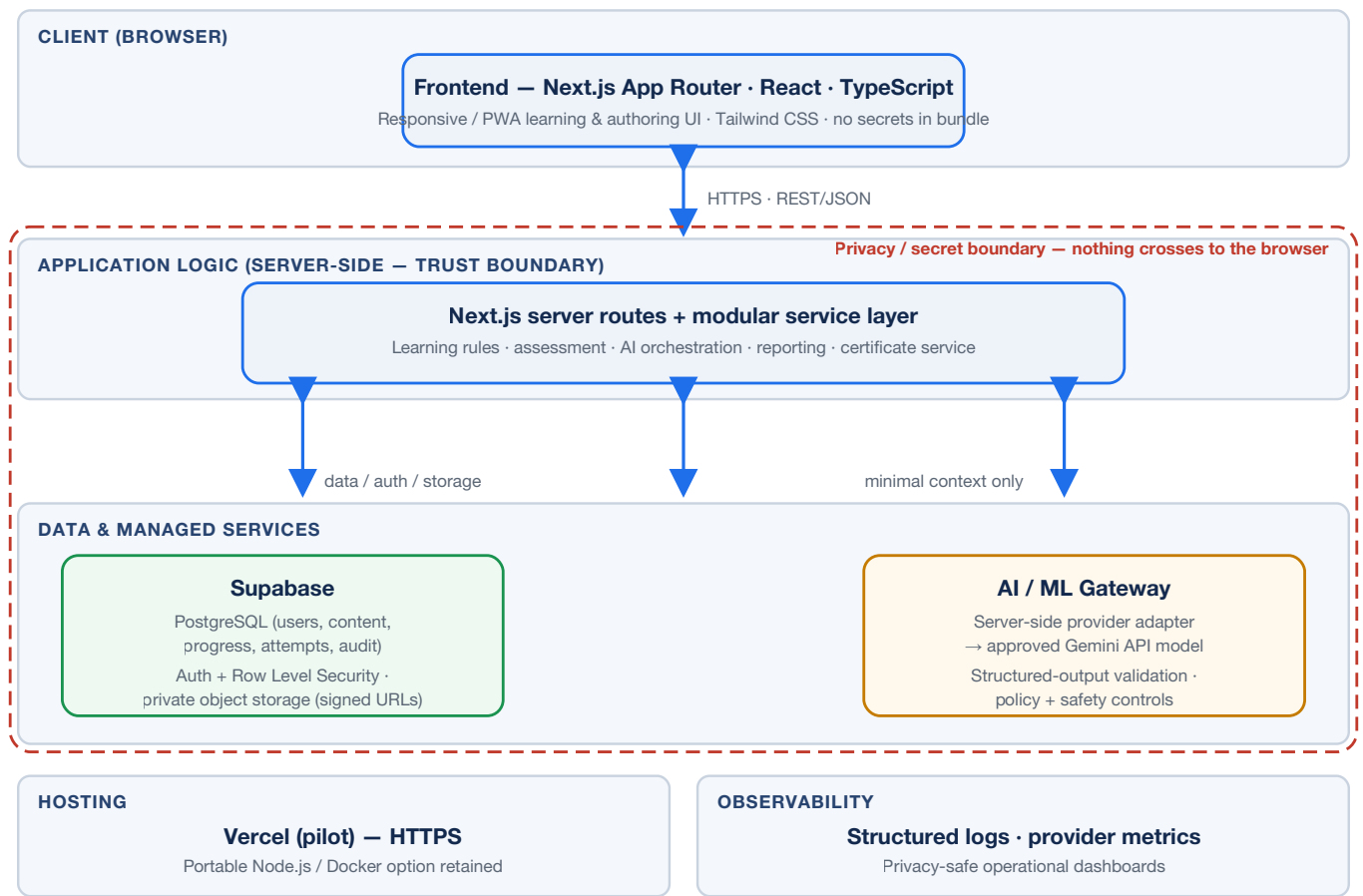


Figure 2. Functional module dependency and shared foundation.

**Progression rule:** A proposed module unlock occurs when all required activities in the preceding module are complete and the participant achieves at least 70% on its quiz. Retries are allowed and the highest valid score is retained.

## 7. High-Level System Architecture

### 7.1 Architecture Diagram



**Figure 3.** Portable, service-oriented web architecture — redrawn to show clear request flow, the server-side trust/privacy boundary, and separation of database, AI gateway, hosting and observability.

### 7.2 Architecture Components

| Component                   | Purpose                                                                           | Proposed Technology                                                                            |
|-----------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Frontend                    | Responsive, accessible learning and authoring interface                           | Next.js App Router, React, TypeScript, Tailwind CSS                                            |
| Backend / Application Logic | Learning rules, assessment, AI orchestration, reporting and certificate service   | Next.js server routes and modular service layer; REST/JSON contracts                           |
| Database                    | Transactional users, content, progress, attempts, configuration and audit records | Managed PostgreSQL through Supabase                                                            |
| AI / ML Gateway             | Prompt evaluation, suggestions and content transformation behind policy controls  | Server-side provider adapter using an approved Gemini API model; structured output validation  |
| Authentication              | Email-based sign-in, sessions, roles and authorization                            | Supabase Auth with Row Level Security; optional IITB SSO in a future phase                     |
| Storage                     | Private uploaded materials, generated artifacts and certificate files             | Supabase private object-storage buckets with signed access                                     |
| Hosting                     | Pilot deployment, HTTPS, scaling and deployment automation                        | Vercel for the pilot; portable Node.js / Docker deployment retained as an institutional option |
| Observability               | Error, latency, usage and security-event visibility                               | Structured application logs, provider metrics and privacy-safe operational dashboards          |

### 7.3 External Services

| Service    | Purpose                                                                                        | Dependency           | Fallback / Control                                                          |
|------------|------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------|
| Gemini API | AI feedback and content transformation                                                         | High for AI features | Rule-based rubric feedback, retry queue and clear service-unavailable state |
| Supabase   | PostgreSQL, authentication and private storage                                                 | High                 | Documented backups and export; service boundaries isolated                  |
| Vercel     | Pilot web hosting and deployment                                                               | Medium               | Deployable as a standard Node.js / Docker application                       |
| GitHub     | Official source control at <a href="#">EAD-Labs/Group-6_2026</a> , review and automated checks | High                 | Local repository mirror and release archives                                |
| NotebookLM | Guided external workflow only; no direct integration in MVP                                    | Low                  | Static step-by-step guide and downloadable checklist                        |

### 7.4 Architecture Principles

- **Secure by default:** authorization is enforced at both service and data layers, and secrets never enter the browser bundle.
- **Provider portability:** application logic depends on stable internal interfaces rather than vendor-specific calls throughout the codebase.
- **Data minimization:** store and transmit only information required for the learning task, progress evidence and approved operations.
- **Versioned evidence:** learning content, rubrics and certificate rules retain versions so historical attempts remain interpretable.
- **Graceful degradation:** non-AI learning remains usable during model-service disruption, while participant input is preserved for retry.
- **Observable and testable:** critical requests carry traceable IDs and the primary journeys are covered by automated and manual checks.

## 8. Data Architecture

### 8.1 Data Flow

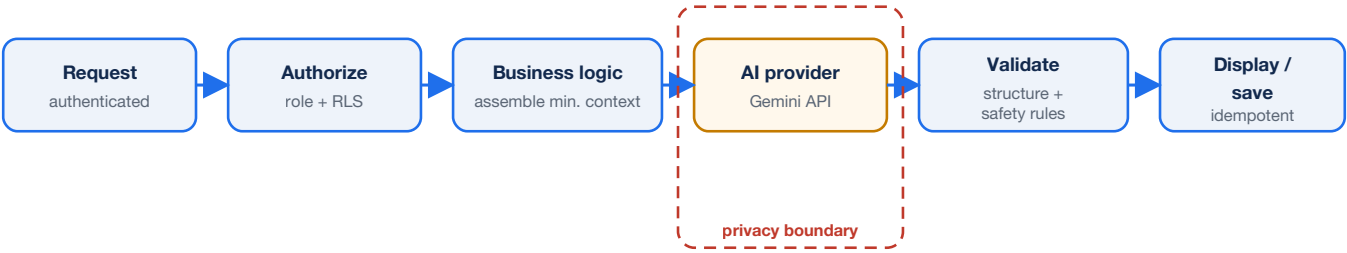


Figure 4. Learning and AI data flow with the privacy boundary around external AI processing.

Authenticated requests are authorized before business logic runs. AI requests are assembled server-side using the minimum required context. Provider responses are validated against expected structure and safety rules before being displayed or saved. Progress writes use idempotent operations so retries do not create duplicate attempts or certificates.

### 8.2 Key Data Entities

| Entity              | Description                               | Key Information                                                           |
|---------------------|-------------------------------------------|---------------------------------------------------------------------------|
| User / Profile      | Authenticated participant or staff member | User ID, email, name, role, consent version, institution context          |
| Cohort              | Programme grouping                        | Cohort ID, dates, facilitator, status                                     |
| Module / Lesson     | Versioned learning structure              | Title, order, prerequisite, content version, publish state                |
| Quiz / Question     | Concept assessment                        | Question, options, rationale, pass threshold, version                     |
| Attempt             | Participant response record               | User, activity, score, response, timestamps, completion state             |
| Prompt Exercise     | Authentic prompt task and rubric          | Scenario, dimensions, criteria, required context                          |
| Prompt Submission   | Participant prompt iteration              | Prompt text, rubric result, feedback, revision link                       |
| Assistant Template  | Reusable teaching-workflow specification  | Purpose, audience, context, constraints, output format                    |
| Resource / Artifact | Teacher-owned upload or generated draft   | Owner, type, storage key, expiry, transformation type                     |
| Progress            | Derived pathway state                     | Completed activities, quiz status, unlocked module, completion percentage |
| Certificate         | Completion evidence                       | Certificate ID, user, issue date, rule version, status                    |
| Audit Event         | Administrative / security trace           | Actor, action, resource, timestamp, outcome                               |

### 8.3 Data Relationships

- A Cohort contains Users; a User has one active role per context and may enrol in multiple cohorts over time.
- A Module contains ordered Lessons and Activities; prerequisite rules determine availability.
- A User submits Attempts and Prompt Submissions; each record references the exact content or rubric version used.
- A User owns Assistant Templates, uploaded Resources and generated Artifacts; access is restricted by owner and role policy.
- Progress is derived from valid activity completion; Certificate issuance records the rule version that established eligibility.

### 8.4 Data Storage, Retention and Exclusions

| Data Category                    | Proposed Storage / Retention                                        | Control                                              |
|----------------------------------|---------------------------------------------------------------------|------------------------------------------------------|
| Account and progress             | Managed PostgreSQL; programme duration plus 12 months of inactivity | Export / delete workflow; client-configurable policy |
| Quiz and prompt practice history | PostgreSQL; aligned with progress retention                         | Owner access; aggregate analytics by default         |

|                             |                                                                                   |                                                   |
|-----------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------|
| Uploaded teaching materials | Private object storage; automatic deletion after 30 days or earlier user deletion | Signed URLs, type / size limits, no student PII   |
| Generated artifacts         | Private object storage; 30 days unless participant explicitly saves               | Ownership checks and user-controlled deletion     |
| Audit / security events     | Restricted log store; 90 days proposed                                            | Admin-only, tamper-evident export where feasible  |
| Certificate register        | PostgreSQL; three years proposed or client policy                                 | Minimal fields; revocation status and audit trail |

**Data deliberately excluded:** Student personally identifiable information, payment data, biometrics, raw passwords, client-side AI keys, and unnecessary demographic profiling. Participants are warned not to upload identifiable student work or confidential institutional material.

## 9. UI / UX Architecture

### 9.1 Screen Inventory

| Screen                 | User                | Purpose                                                                                   |
|------------------------|---------------------|-------------------------------------------------------------------------------------------|
| Sign-in & Onboarding   | All                 | Authenticate, accept privacy notice, set teaching context and explain safe-use boundaries |
| Learning Dashboard     | Participant         | Resume learning, view modules, requirements, progress and next recommended action         |
| Module Overview        | Participant         | Understand outcomes, prerequisite status and activity sequence                            |
| Lesson Player          | Participant         | View micro-video / readings with transcript, key points and completion state              |
| Concept Quiz           | Participant         | Answer questions, receive explanations and retry                                          |
| Prompt Sandbox         | Participant         | Draft, evaluate, improve and compare prompts                                              |
| Assistant Builder      | Participant         | Define and save a reusable teaching-assistant template                                    |
| Content Transformer    | Participant         | Upload / paste material, choose output, review and export a draft                         |
| Progress & Certificate | Participant         | Review evidence, incomplete requirements and download certificate                         |
| Content Studio         | Facilitator / Admin | Draft, preview, version and publish programme content                                     |
| Cohort Insights        | Facilitator / Admin | Review aggregate progress, common errors and export summaries                             |
| Administration         | Admin               | Manage roles, cohorts, policies, prerequisites, certificates and audit events             |

### 9.2 Navigation Structure

**Participant:** Sign-in → Dashboard → Module → Lesson / Quiz / Practice → Feedback → Next Activity → Progress / Certificate. **Staff:** Sign-in → Content Studio or Cohort Insights → Draft / Preview / Publish or Review / Export. Navigation remains shallow, exposes one clear primary action per screen, and never relies on colour alone.

### 9.3 Conceptual Wireframes

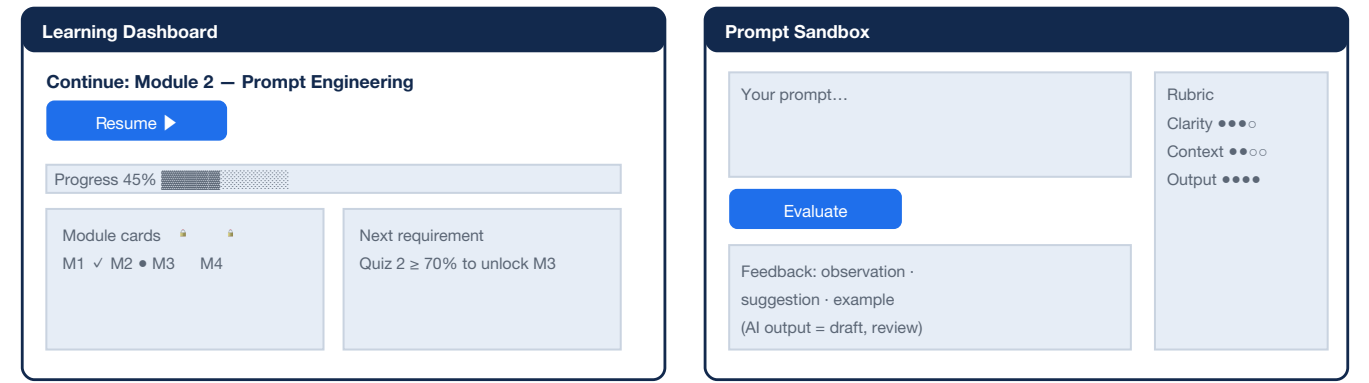


Figure 5. Conceptual dashboard and prompt-sandbox wireframes; visual design remains subject to client review.

#### Screen-Level UX Rules

- Every screen presents a visible title, current location and one dominant next action.
- Locked modules explain exactly what must be completed; disabled controls are not used without an explanation.
- AI feedback separates observation, suggestion and example so participants can decide how to revise.
- Long-running AI actions show progress, allow cancellation where feasible and preserve the participant's input on failure.
- Videos include captions / transcripts; quizzes and rubric results are keyboard and screen-reader operable.

## 10. Technology Stack

**Status:** The stack below is a proposed draft baseline chosen for type safety, rapid delivery, managed security controls and deployment portability. Final service accounts, regions and model selection require client approval.

### 10.1 Frontend

| Item              | Proposed Choice                                   | Rationale                                                                |
|-------------------|---------------------------------------------------|--------------------------------------------------------------------------|
| Platform          | Responsive web / PWA-compatible                   | One codebase across laptops, tablets and phones; no app-store dependency |
| Framework         | Next.js App Router with React                     | Server / client composition, routing and portable Node.js deployment     |
| Language          | TypeScript                                        | End-to-end type safety and maintainable contracts                        |
| Styling           | Tailwind CSS plus accessible component primitives | Consistent responsive system with controlled tokens                      |
| Client validation | Zod-compatible schemas                            | Shared, explicit validation for forms and structured AI output           |

### 10.2 Backend

| Item                | Proposed Choice                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------|
| Framework / Runtime | Next.js server routes and modular TypeScript service layer on Node.js                                  |
| API architecture    | REST/JSON endpoints with versioned schemas; server-only AI and privileged data access                  |
| Authorization       | Role policy plus PostgreSQL Row Level Security; deny by default                                        |
| Background work     | Idempotent jobs for AI retries, certificate generation and cleanup                                     |
| Testing             | Unit / component tests, API integration tests and browser-based end-to-end tests for critical journeys |

### 10.3 Database and Storage

**Database:** Supabase-managed PostgreSQL for transactional and reporting data.

**File storage:** Private Supabase buckets; signed, short-lived access; lifecycle deletion for temporary materials.

**Migration strategy:** Version-controlled SQL migrations, seeded non-sensitive test data and documented backup / export procedure.

### 10.4 AI / ML Components

| Item              | Design                                                                                                                   |
|-------------------|--------------------------------------------------------------------------------------------------------------------------|
| Model / Service   | Approved Gemini API model accessed through a provider-neutral server adapter; exact model selected during implementation |
| Purpose           | Prompt rubric feedback, revision suggestions and teacher-controlled content transformation                               |
| Inference         | Cloud; no AI key or privileged prompt template is sent to the browser                                                    |
| Structured output | Schema-validated JSON for rubric dimensions, feedback and warnings before UI rendering                                   |
| Safety            | Provider safety settings, prompt-injection resistance, input limits, output checks and visible human-review reminder     |
| Fallback          | Deterministic rubric checklist and saved draft; retry option when the external AI service is unavailable                 |

### 10.5 Infrastructure and Delivery

| Area                | Proposed Choice                                                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pilot hosting       | Vercel with HTTPS and environment-managed secrets                                                                                                                                     |
| Portable deployment | Standard Node.js server or Docker image for an approved institutional host                                                                                                            |
| Version control     | Official GitHub repository: <a href="https://github.com/EAD-Labs/Group-6_2026">github.com/EAD-Labs/Group-6_2026</a> ; protected main branch, feature branches and pull-request review |
| Work tracking       |                                                                                                                                                                                       |

|              |                                                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Jira project with epics for Recovery/Design, Delivery Modules 1–3, Integration/Release and Field Study; every weekly deliverable has an owner, due date, acceptance evidence and status |
| CI/CD        | Automated lint, type-check, test, accessibility smoke check and deployment preview                                                                                                      |
| Environments | Local, preview / staging and production with separate credentials and databases                                                                                                         |

## 10.6 Repository and Jira Working Agreement NEW

- **Repository of record:** [https://github.com/EAD-Labs/Group-6\\_2026](https://github.com/EAD-Labs/Group-6_2026). The repository is owned by EAD-Labs so project history remains available after the course.
- **Contribution evidence:** all four students work through named accounts, feature branches and reviewed pull requests; direct commits to main are avoided.
- **Jira linkage:** branch names, commits and pull requests reference the Jira issue key. An issue is Done only when its acceptance evidence is linked.
- **Security:** API keys, tokens and production credentials are never committed; an `.env.example` documents required variables.
- **Weekly review:** Friday status uses the Jira board, staging URL, merged pull requests and test evidence rather than verbal progress alone.



## 11. Non-Functional Requirements

### 11.1 Performance and Reliability

| NFR                    | Draft Target                                                                                                    | Verification                        |
|------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Initial dashboard load | ≤ 3 seconds on a representative 4G connection for cached / static assets                                        | Browser performance test            |
| Non-AI interaction     | 95th percentile ≤ 2 seconds under pilot load                                                                    | API load test                       |
| AI feedback            | Target ≤ 15 seconds; hard timeout ≤ 30 seconds with recoverable error                                           | Instrumented integration test       |
| Concurrent users       | <b>REVISED</b> Up to 25 concurrent authenticated users for the 10-user pilot architecture (headroom for growth) | Staged load test                    |
| Availability           | 99.5% during scheduled pilot periods, excluding announced maintenance and external-service outage               | Provider and application monitoring |
| Recovery               | Documented daily backup / export and restore rehearsal before field study                                       | Restore evidence                    |

### 11.2 Accessibility

- Target WCAG 2.2 Level AA for participant-facing critical journeys.
- All functionality operable by keyboard with visible focus and logical focus order.
- Semantic headings, landmarks, labels, error association and meaningful alternative text.
- Captions and transcripts for instructional video; no essential information conveyed through colour alone.
- Minimum contrast and responsive text validated through automated checks and manual review.

### 11.3 Security

- TLS for all network communication; secure, HTTP-only session handling where applicable.
- Server-side secrets, least-privilege service roles and Row Level Security for user-owned data.
- Role checks at both API and database layers; administrative actions logged.
- Rate limits, request-size limits, input validation and safe error messages.
- Private file buckets, signed URLs, file-type / size validation and deletion lifecycle.
- Dependency, static-analysis and critical-flow security checks before production release.

### 11.4 Privacy and Responsible AI

| Control                | Requirement                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| PII collected          | Name, email, role, institution / cohort context and learning records only as required                                              |
| Student data           | Not required and explicitly prohibited in uploads or prompts                                                                       |
| Data minimization      | Send only task-relevant context to the AI provider; do not include account metadata unless necessary                               |
| Transparency           | Explain when AI is used, what is sent, limitations, retention and how the participant can delete material                          |
| Human oversight        | All generated educational content is a draft; the teacher remains responsible for accuracy, bias, suitability and copyright review |
| Analytics              | Aggregate by default; no public leaderboard or unnecessary comparison between participants                                         |
| Third-party processing | Requires approved service configuration, contract / terms review and a client-approved privacy notice before pilot use             |

### 11.5 Compatibility

**Supported devices:** Current laptops / desktops, tablets and smartphones with a viewport of at least 360 CSS pixels.

**Supported browsers:** Current and previous major releases of Chrome, Edge, Firefox and Safari at release time; exact support matrix recorded during UAT.

### 11.6 Offline Requirements

The MVP is online-first. The application shell, previously opened text lessons and unsent draft input may be cached where safe. Authentication, AI evaluation, quiz submission, progress synchronization, administration and certificate issuance require connectivity. The interface must clearly indicate offline state and never imply that unsynchronized work has been submitted.

## 12. Assumptions & Constraints

---

### 12.1 Assumptions

1. The Client will supply or approve final learning content, micro-videos, quiz banks, prompt rubrics and certificate wording.
2. Participants have an email address, a modern browser and periodic access to stable internet; English is the MVP interface language.
3. **REVISED** The initial pilot includes **10 participants** (school teachers and educators) and does not require more than 25 concurrent authenticated sessions.
4. The Client will approve an AI provider, service account, usage budget, privacy notice and acceptable-use guidance before live AI features are enabled.
5. Participants use teacher-owned or non-sensitive materials and will not submit identifiable student information.
6. Completion requires all mandatory activities and at least 70% on each module quiz, with retries; this rule is subject to client confirmation.
7. The application supports professional learning and formative practice; it is not a high-stakes assessment, autonomous grading or decision-making system.

### 12.2 Constraints

1. The academic delivery window runs from August to November 2026 and the core team consists of four students.
2. External AI, hosting and managed-data services may impose availability, quota, pricing and policy constraints outside the Student Team's control.
3. AI output may be inaccurate, biased or unsuitable; safeguards reduce but cannot eliminate this risk.
4. Variable connectivity and device capability may affect video playback and AI response time.
5. Content, API access, approval and feedback delays may move dependent milestones.
6. Native mobile applications, enterprise integrations and multilingual content are outside the MVP unless scope is changed by mutual agreement.

## 13. Project Scope

### 13.1 In Scope – Minimum Viable Product

| Scope Area             | Included Capability                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| Access                 | Teacher authentication, secure sessions, onboarding, participant profile and role policy                          |
| Learning pathway       | Four sequenced learning modules with micro-videos, readings, activity navigation and prerequisite-based unlocking |
| Assessment             | Auto-graded concept quizzes, explanations, retries, completion thresholds and progress tracking                   |
| Prompt practice        | Interactive sandbox, rubric-based AI evaluation, improvement suggestions and attempt comparison                   |
| Reusable AI workflows  | Custom AI Assistant (Gem-style) specification builder with save / edit / export                                   |
| Content transformation | Teacher-controlled transformation of material into summaries, quizzes and study guides with review checklist      |
| NotebookLM             | Guided workflow, examples and safety checklist; no direct API integration                                         |
| Completion             | Progress dashboard, eligibility calculation, completion certificate and certificate register                      |
| Administration         | Content draft / preview / publish, module rules, cohorts, users / roles and aggregate reports                     |
| Quality                | Responsive design, accessibility target, security / privacy safeguards, tests, deployment and documentation       |

### 13.2 Delivery Modules & Module-Level Testing NEW

Per client feedback, the build is split into **three delivery modules**. Each is developed, deployed to staging and **tested separately** (unit, integration and a module-level UAT / demo) before the next module begins. A per-module test report is shared with the Client at each module handover.

| Delivery Module                                   | System Modules Covered                                                   | Outcome Delivered & Tested Separately                                                                                                                                 |
|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Module 1</b><br>Foundation & Learning Core     | M1 Identity & Access · M2 Learning Experience · M3 Assessment & Progress | Authenticated participants complete sequenced lessons and quizzes (retries), with prerequisite gating and progress state. Tested end-to-end before Module 2 starts.   |
| <b>Module 2</b><br>AI Practice & Workflows        | M4 Prompt Practice · M5 AI Workflow Tools                                | Rubric-driven prompt sandbox, reusable assistant builder, content transformer and NotebookLM guide with safeguards. Tested separately, including AI-failure handling. |
| <b>Module 3</b><br>Certification & Administration | M6 Certification · M7 Administration & Insights                          | Eligibility, certificate + register, content lifecycle, cohorts, roles, aggregate reports and audit. Tested separately.                                               |
| <b>Integration &amp; Release</b>                  |                                                                          | Cross-module end-to-end journeys, accessibility, performance, security and the complete project presentation — completed by <b>15 Oct 2026</b> .                      |

### 13.3 Out of Scope for the MVP

- Native Android or iOS applications and full offline course completion.
- Learning-analytics warehouse, predictive / adaptive pathways or automated course recommendations.
- Personalized conversational AI coach beyond scoped feedback workflows.
- Teacher social network, peer prompt marketplace or collaborative communities.
- Institutional administrator dashboard across multiple organisations, SSO / LMS integration and institutional certificate management.
- Badge economy, gamification, multilingual interface and large-scale public deployment.
- Processing of identifiable student records, autonomous classroom decisions, high-stakes assessment or unsupervised publication of AI output.
- Long-term maintenance, paid production infrastructure and legal / compliance certification beyond the ET 617 project unless separately agreed.

**Change control:** Any functionality not explicitly listed as in scope requires a documented decision covering value, effort, privacy / safety impact, timeline and acceptance evidence before implementation.

## 14. Acceptance Criteria

The application is functionally complete when every Must-have criterion below is demonstrated in the approved deployment and no open Critical defect prevents a primary journey.

| ID    | Area                                                                                                  | Acceptance Criterion                                                                                                                                    | Evidence                 |
|-------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| AC-01 | Access                                                                                                | A participant can authenticate, sign out and resume the correct account state; unauthorized role actions are blocked.                                   | End-to-end test          |
| AC-02 | Learning                                                                                              | Four modules present approved videos / readings / activities in the agreed order and save completion state.                                             | Content checklist + demo |
| AC-03 | Quiz                                                                                                  | Quizzes score automatically, show explanations, allow retry and retain a valid attempt history.                                                         | Test cases + demo        |
| AC-04 | Unlocking                                                                                             | A module unlocks only after required prior activities and the approved quiz threshold are complete.                                                     | Boundary tests           |
| AC-05 | Prompt sandbox                                                                                        | A participant can submit, receive dimension-level feedback, revise and compare at least two prompt attempts.                                            | Scenario demo            |
| AC-06 | AI resilience                                                                                         | AI timeout or provider error preserves input and provides deterministic guidance plus retry without corrupting progress.                                | Failure-injection test   |
| AC-07 | Assistant builder                                                                                     | A participant can create, edit, save and export a reusable assistant specification.                                                                     | Scenario demo            |
| AC-08 | Transformer                                                                                           | A participant can supply an allowed resource, select an output type, receive an editable draft and delete the resource.                                 | Scenario demo            |
| AC-09 | Progress                                                                                              | Dashboard percentage and outstanding requirements match underlying activity records.                                                                    | Data reconciliation      |
| AC-10 | Certificate                                                                                           | Eligible participants can download a certificate; ineligible participants see exact missing requirements.                                               | Eligibility tests        |
| AC-11 | Administration                                                                                        | Authorized staff can draft, preview and publish content and configure cohort / module settings.                                                         | Role-based demo          |
| AC-12 | Reporting                                                                                             | Facilitator can view / export assigned-cohort aggregate completion without routine access to private prompts / uploads.                                 | Permission test          |
| AC-13 | Responsive UX                                                                                         | Primary journeys work at 360 px mobile, tablet and desktop widths without clipping or horizontal page scroll.                                           | Visual regression / UAT  |
| AC-14 | Accessibility                                                                                         | Critical journeys pass agreed automated checks and a manual keyboard / screen-reader smoke review against WCAG 2.2 AA target.                           | Audit report             |
| AC-15 | Performance                                                                                           | Measured response targets in Section 11.1 are met under the agreed pilot profile or deviations are accepted in writing.                                 | Performance report       |
| AC-16 | Privacy                                                                                               | Consent, no-student-PII warning, private-file access and deletion / expiry behavior are implemented and documented.                                     | Privacy checklist        |
| AC-17 | Security                                                                                              | No open Critical / High finding in the agreed pre-release security checklist; secrets are absent from client code / repository.                         | Security review          |
| AC-18 | Delivery                                                                                              | Production / staging deployment, source code, setup guide, user guide, HLD, test summary and presentation are handed over.                              | Handover checklist       |
| AC-19 | Module testing<br> | Each of the three delivery modules passes its own unit, integration and UAT / demo before the next module begins; a per-module test report is provided. | Per-module test reports  |

### 14.1 Defect Severity and Sign-off

- **Critical:** security / privacy breach, data loss or inability to complete a primary journey — must be resolved before release.
- **High:** major function fails with no acceptable workaround — must be resolved or explicitly waived by the Client.
- **Medium / Low:** limited impact or cosmetic issue — may be scheduled with an agreed owner and date.
- Client UAT feedback should be consolidated within each module's review window so the Student Team can assess and incorporate it before the next module.

## 15. Development Dependencies

**Re-baselined on 2 Sep 2026:** The client provides direction, review, approvals and suitable examples; the Student Team curates or creates the learning materials and implementation. Placeholder content may be used only until client confirmation.

| Dependency / Decision                                                                         | Owner         | Required By | Status / Impact                                                             |
|-----------------------------------------------------------------------------------------------|---------------|-------------|-----------------------------------------------------------------------------|
| Confirm PromptShala name, primary teacher persona and four learning outcomes                  | Client + Team | 4 Sep 2026  | Name selected by Team; persona/ outcomes require consolidated client review |
| Curated or team-created sample content for AI introduction and prompt practice                | Student Team  | 6 Sep 2026  | Required for Module 1 prototype; Creative Commons or embedded YouTube only  |
| Confirm LEARN-LM / CRAFT-style prompt rubric and example classroom scenarios                  | Client + Team | 8 Sep 2026  | Blocks final validation rules and test cases                                |
| Privacy notice, acceptable-use wording and API-key explanation                                | Client + Team | 10 Sep 2026 | Required before any external participant use                                |
| AI implementation decision: lightweight rules/model first; teacher API key only when required | Client + Team | 18 Sep 2026 | Blocks production configuration for Module 2                                |
| Non-sensitive sample syllabus/worksheet for transformation testing                            | Client        | 25 Sep 2026 | Required for Module 2 transformer test evidence                             |
| Hosting/domain credentials or approval to use team staging                                    | Client        | 5 Oct 2026  | Staging can continue independently until release                            |
| Pilot participant list and schedule (10 teachers/educators)                                   | Client        | 10 Oct 2026 | Required for field-study planning                                           |
| Consolidated final UAT feedback and release decision                                          | Client        | 13 Oct 2026 | Required before final presentation on 15 Oct                                |

## 16. Open Questions & Decision Log

Version 1.2 records the signed Version 1.1 baseline and the client directions from 21 and 28 Aug. Items marked Confirmed are current working decisions; remaining items require consolidated client confirmation.

| ID   | Question / Decision                   | Proposed Decision                                                                                                        | Owner         | Date / Status                   |
|------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------|
| D-01 | Delivery platform                     | Responsive web application; PWA-compatible; no native app in MVP                                                         | Client + Team | 14 Aug — Proposed               |
| D-02 | End-user roles                        | Participant, Facilitator / Content Manager and Administrator                                                             | Client + Team | 14 Aug — Proposed               |
| D-03 | Completion rule                       | Required activities plus $\geq 70\%$ on each module quiz; retries allowed                                                | Client        | 14 Aug — Proposed               |
| D-04 | Technology baseline                   | Next.js / TypeScript, Supabase data services, server-side Gemini adapter, portable hosting                               | Team + Client | 14 Aug — Proposed               |
| D-05 | Pilot capacity<br><b>REVISED</b>      | Initial pilot of 10 participants (school teachers & educators); test for up to 25 concurrent authenticated users         | Client        | 17 Aug — Proposed               |
| D-06 | Privacy boundary                      | No student PII; private uploads; 30-day temporary-file retention; aggregate analytics                                    | Client        | 14 Aug — Proposed               |
| D-07 | Scope boundary                        | Client-brief MVP is in scope; listed future enhancements are out of scope                                                | Client + Team | 14 Aug — Proposed               |
| D-08 | Client confirmation                   | Prof. Ramkumar Rajendran and Aditya will both sign as Client representatives                                             | Client        | 14 Aug —<br>Confirmed by user   |
| D-09 | Delivery structure<br><b>NEW</b>      | Project delivered as three separately built and separately tested modules; complete project presentation by 15 Oct 2026  | Client + Team | 17 Aug — Proposed               |
| D-10 | Deployment context<br><b>NEW</b>      | Intended audience is school teachers and other educators (not a single campus); CET/IITB is client and pilot coordinator | Client + Team | 17 Aug — Proposed               |
| D-11 | Product identity <b>NEW</b>           | <b>PromptShala</b> — Practical AI Literacy Studio for Educators                                                          | Student Team  | 2 Sep — Confirmed               |
| D-12 | Repository of record<br><b>NEW</b>    | <a href="#">EAD-Labs/Group-6_2026</a> ; Jira tracks all weekly work and evidence                                         | Client + Team | 2 Sep — Confirmed               |
| D-13 | Recovery schedule<br><b>NEW</b>       | Re-baseline from 2 Sep with weekly evidence; separately test all three delivery modules; retain 15 Oct presentation      | Client + Team | 2 Sep — Proposed                |
| D-14 | Teacher-specific design<br><b>NEW</b> | Scenarios, rubrics, safety and outputs are designed for beginner school teachers, not generic professionals              | Client + Team | 28 Aug —<br>Confirmed direction |

**Approval action:** Before signing, the Client should confirm the proposed items in D-01 through D-14 or record an alternative decision in this section. Any change must be reflected in the relevant HLD section and milestone plan.

## 17. Project & HLD Agreement

The Student Team and the Client confirm that they have reviewed and discussed the project requirements, proposed solution, user personas, workflows, system design, project scope, technical approach and deliverables described in this High-Level Design (HLD) Document. The HLD, together with this Agreement, forms the agreed basis for development of PromptShala within the ET 617 academic project.

### 17.1 Agreement on HLD

- The approved HLD represents the current agreed understanding of the project.
- The HLD defines intended users, user journeys, core functionality, system architecture, technology approach, data safeguards, scope, assumptions, constraints and acceptance criteria.
- The Student Team will use the approved HLD as the basis for design, development, testing and delivery.
- The Client will review the HLD and provide consolidated feedback or approval within the agreed schedule.
- A significant change to scope, privacy, functionality, acceptance criteria or external dependencies will be discussed before implementation.
- Approved changes may affect the milestone plan and will be incorporated only after mutual agreement and an updated decision record.

### 17.2 Agreement on Responsibilities

#### Student Team Responsibilities

- Complete agreed development work and demonstrate milestone deliverables.
- Deliver and separately test each of the three modules, providing a per-module test report.
- Maintain source code in the official EAD-Labs GitHub repository, track weekly work in Jira, and retain technical documentation, test evidence and the decision log.
- Apply reasonable security, privacy, accessibility and responsible-AI safeguards described in this HLD.
- Incorporate reasonable, consolidated feedback received within the agreed review period.
- Communicate technical risks, dependency issues or timeline impacts as soon as reasonably possible.
- Provide the final deliverables identified in Section 17.4.

#### Client Responsibilities

- Provide accurate content, non-sensitive sample data, service access, credentials and approvals required for development.
- Nominate an appropriate representative for meetings and milestone review.
- Review each module deliverable within the specified feedback period and provide clear, consolidated feedback.
- Approve the AI service, privacy notice, acceptable-use guidance and pilot conditions before live participant use.
- Communicate major requirement changes as early as possible and participate in scope / timeline decisions.

### 17.3 Timeline Dependencies

The development timeline depends on timely participation, content, access, approvals and feedback from both parties. If a required Client dependency is delayed, the corresponding module milestone may be revised. If the Student Team identifies a technical or implementation issue that materially affects scope, quality or timeline, it will communicate the issue promptly with proposed options.

### 17.4 Final Deliverables and Project Timeline

#### Final Deliverables

- **Executable Application:** final working educational application / prototype in the approved environment.
- **Source Code:** complete source in [EAD-Labs/Group-6\\_2026](#), configuration template, commit/PR history and release tag / archive.
- **Per-Module Test Reports:** separate unit / integration / UAT evidence for each of the three delivery modules.
- **User Documentation:** participant and administrator instructions for the delivered functions.
- **Technical Documentation:** HLD, architecture, data model, setup / deployment guide and test summary.
- **Project Presentation:** final demonstration and presentation of process, product and outcomes — **delivered by 15 Oct 2026**.
- **Field Study & Analysis:** approved study materials, data summary and analysis presentation, subject to course and client requirements.

#### Week-wise Recovery and Delivery Schedule **REVISED**

**Status at re-baseline (2 Sep 2026):** HLD revisions and preliminary concepts exist, but the official repository has no project code and a complete approved UI/UX prototype has not been demonstrated. The plan below is a recovery schedule; the 15 Oct presentation date remains fixed.

| Week / Dates               | Focus                               | Required Output                                                                                                                                                     | Evidence / Separate Test                                                      | Review |
|----------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|
| <b>Recovery</b><br>2–6 Sep | Baseline, persona, design and setup | HLD v1.2; PromptShala identity; detailed teacher persona; finalized user flow/wireframes; Jira board; repository scaffold, README, branching rules and staging plan | Approved documents, Jira backlog, visible GitHub commits/PR, clickable design | 6 Sep  |

|                                    |                                           |                                                                                                                                   |                                                                              |                            |
|------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------|
|                                    |                                           |                                                                                                                                   | or complete wireframe walkthrough                                            |                            |
| <b>Week 1</b><br>7–13 Sep          | Module 1A — Access & learning shell       | Sign-in, onboarding, teacher profile, dashboard, four-module pathway, lesson/video/transcript shell and saved progress            | Staging demo; access/role tests; responsive smoke test                       | 13 Sep                     |
| <b>Week 2</b><br>14–20 Sep         | Module 1B — Assessment & progress         | Auto-graded quiz, explanations, retries, 70% gate, unlock logic and progress calculation                                          | <b>Module 1 test report:</b> unit, integration, end-to-end/UAT evidence      | 20 Sep; feedback by 21 Sep |
| <b>Week 3</b><br>21–27 Sep         | Module 2A — Prompt practice               | Teacher scenarios, LEARN-LM/CRAFT-style rubric, prompt submission, missing-dimension feedback, revision and attempt comparison    | Rubric test cases; safe fallback; prompt-practice staging demo               | 27 Sep                     |
| <b>Week 4</b><br>28 Sep–4 Oct      | Module 2B — Reusable workflows            | Gem-style assistant specification builder, save/edit/export, NotebookLM guide and safeguarded material transformation             | <b>Module 2 test report:</b> AI failure, privacy and workflow tests          | 4 Oct; feedback by 5 Oct   |
| <b>Week 5</b><br>5–9 Oct           | Module 3 — Certification & administration | Eligibility, certificate/register, content draft/preview/publish, roles, cohorts and aggregate progress                           | <b>Module 3 test report:</b> eligibility, RBAC, reporting and audit tests    | 9 Oct; feedback by 10 Oct  |
| <b>Week 6</b><br>10–13 Oct         | Integration, QA and UAT                   | Cross-module journeys, defect fixes, accessibility, performance, security/privacy checks, user/admin guides and release candidate | Integrated test summary, UAT checklist, defect register and release decision | 13 Oct                     |
| <b>Release</b><br>14–15 Oct        | Handover & presentation                   | Final application, source/release tag, setup guide, HLD, test reports, user docs and complete project presentation                | Final handover checklist and live demonstration                              | <b>15 Oct</b>              |
| <b>Field Study</b><br>16 Oct–6 Nov | 10-user pilot and analysis                | Approved pilot, data collection, analysis and findings presentation                                                               | Study evidence and privacy-safe analysis                                     | 6 Nov                      |

## 17.5 Revision and Signature Continuity

**Signed baseline retained:** The final page of signed Version 1.1 is appended unchanged as the last page of this revision, as requested by the Student Team. It records approval history for the earlier baseline. Version 1.2 changes are documented in the revision summary, decision log and week-wise schedule and remain subject to consolidated client review.



|                       |              |                                                                                                       |                                                                 |              |
|-----------------------|--------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------|
| <b>Module 2</b>       | 6–19 Sep     | Prompt sandbox, assistant builder, transformer, NotebookLM guide                                      | Delivery Module 2 on staging + Module 2 test report             | 22 Sep       |
| <b>Module 3</b>       | 20 Sep–3 Oct | Certification and administration / insights                                                           | Delivery Module 3 on staging + Module 3 test report             | 6 Oct        |
| Integration & Release | 4–15 Oct     | End-to-end integration, accessibility, performance, security;<br><b>complete project presentation</b> | Final application, source, docs & presentation <b>by 15 Oct</b> | 13 Oct (UAT) |
| Field Study           | 16 Oct–6 Nov | Pilot with 10 participants, data collection and analysis                                              | Field-study & data-analysis presentation                        | 6 Nov        |

## 17.5 Mutual Confirmation

**Mutual confirmation:** By signing below, both parties confirm that they have reviewed and discussed the project requirements and HLD, agree to use the approved HLD as the development baseline, and commit to timely collaboration, access, feedback and approvals necessary to pursue the agreed deliverables.

This Draft becomes the approved Version 1.1 baseline only when the approval decision, approval date and required signatures are completed. Digital-signature cells are intentionally blank. Each signer should insert one digital-signature block that includes the signing date in the same cell.

### Digital Signatures — Student Team

| Student Name (Roll No.)    | Digital Signature & Date                                                         |
|----------------------------|----------------------------------------------------------------------------------|
| Darshan Sonawane (23b1541) | Digital Signature<br>Darshan Avinash Sonawane (23b1541)<br>17-Aug-26 09:37:33 PM |
| Vishal Patel (23b0745)     | Digital Signature<br>Vishal Chandra Patel (23b0745)<br>17-Aug-26 10:29:10 PM     |
| Ashok Kumar (23b0628)      | Digital Signature<br>Ashok Kumar (23b0628)<br>18-Aug-26 12:57:46 AM              |
| Raghuram Gundi (23b1534)   | Digital Signature<br>Gundi Raghuram (23b1534)<br>18-Aug-26 05:18:44 AM           |

### Client Representatives

| Client Representative    | Organization                                        | Digital Signature & Date                                                    |
|--------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------|
| Prof. Ramkumar Rajendran | Centre for Educational Technology (CET), IIT Bombay | Digital Signature<br>Ramkumar Rajendran (10001783)<br>18-Aug-26 10:23:15 AM |
| Aditya                   | Centre for Educational Technology (CET), IIT Bombay |                                                                             |

### HLD Approval

| Approval Field        | Value                                           |
|-----------------------|-------------------------------------------------|
| HLD Document Version  | 1.1 (Revision 1)                                |
| Document Status       | Draft — pending approval                        |
| HLD Approval Decision | Approved / Approved with changes / Not approved |
| HLD Approval Date     | _____                                           |